



SYDNEY BOYS HIGH
MOORE PARK, SURRY HILLS

JUNE 2008
TASK #3
YEAR 12

Mathematics

General Instructions:

- Reading time—5 minutes.
- Working time—2 hours.
- Write using black or blue pen.
- Board approved calculators may be used.
- All necessary working should be shown in every question if full marks are to be awarded.
- Marks may NOT be awarded for messy or badly arranged work.
- Start each NEW section in a separate answer booklet.
- Hand in your answer booklets in 3 sections:
Section A(Questions 1 and 2),
Section B(Questions 3 and 4),
Section C(Questions 5 and 6),

Total marks—90 Marks

- Attempt questions 1–6.
- All questions are of equal value.

Examiner: Mr D. Hespe

This is an assessment task only and does not necessarily reflect the content or format of the Higher School Certificate.

Section A

Marks

Question 1 (15 marks)

(a) Find primitives of the following differential functions:

(i) $f'(x) = 2x - 3$

1

(ii) $f'(x) = x^2 - 2x + 1$

1

(iii) $f'(x) = 0$

1

(iv) $f'(x) = (3x + 1)(x - 2)$

2

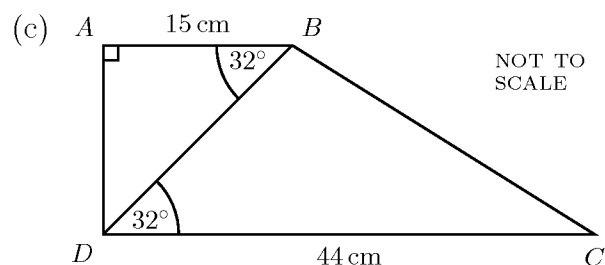
(b) $ABCD$ is a quadrilateral where $A = (-3, 0)$, $B = (0, 4)$, $C = (5, 4)$, and $D = (2, 0)$.

(i) Show that $ABCD$ is a rhombus.

3

(ii) Prove that its diagonals are perpendicular bisectors of one another.

2



(i) Prove that $ABCD$ is a trapezium.

2

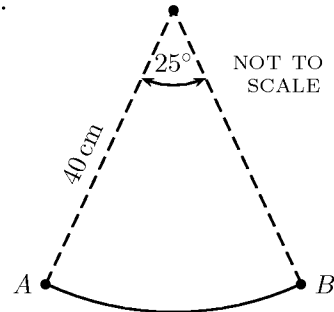
(ii) Calculate the length of BC to a suitable degree of accuracy.

3

Question 2 (15 marks)

- (a) A pendulum 40 cm long swings through an angle of 25° .
Find the length of the arc AB
correct to the nearest millimetre.

2



- (b) Find the exact value of

(i) $\sin \frac{\pi}{4}$

1

(ii) $\cot \frac{4\pi}{3}$

1

(iii) any angles where $\cos \theta = \frac{\sqrt{3}}{2}$, $-\pi \leq \theta \leq \pi$.

2

- (c) Find

(i) $\frac{d}{dx} \sin(3x + \pi)$

1

(ii) $\frac{d}{dx} \log_e(x^2 - x + 2)$

1

(iii) $\frac{d}{dx} (e^{2x} - e^x - e^{-x})$

1

(iv) $\frac{d}{d\theta} (\ln \tan \theta)$

2

- (d) Calculate, correct to 2 decimal places:

(i) $\sin 2$

1

(ii) $\ln 17$

1

- (e) Find the second derivative of e^{x^2} .

2

Section B

(Use a separate writing booklet.)

Marks

Question 3 (15 marks)

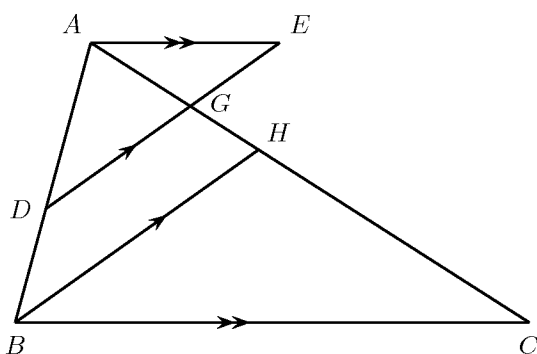
(a) Show that

$$(i) \int_0^{\pi/4} \cos 2x \, dx = 0.5, \quad \boxed{2}$$

$$(ii) \int_2^7 \left| 3 - \frac{x}{2} \right| \, dx = 4.25. \quad \boxed{3}$$

(b) Find the equations of the circles which pass through the points $(3, 0)$ and $(12, 0)$ and are tangent to the y -axis. $\boxed{4}$

(c)



In the diagram (NOT TO SCALE) AE is parallel to BC and $AE = \frac{2}{7}BC$. The point D on AB is such that $BD = \frac{3}{7}BA$. The line DE meets AC at G and the line through B parallel to DE meets AC at H .

(i) Prove that $\widehat{AGE} = \widehat{BHC}$. $\boxed{3}$

(ii) Prove that $\triangle AEG$ is similar to $\triangle CBH$. $\boxed{3}$

Question 4 (15 marks)

- (a) Find in radians all solutions to $\sin x = 0.8$ with $0 \leq x \leq 2\pi$. Give your answers to 4 decimal places. 2

- (b) Evaluate $\int_0^{\pi/4} \sec^2 x \, dx$ by

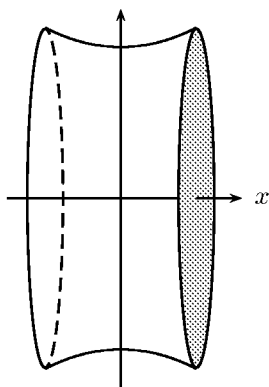
(i) direct integration, 2

(ii) the trapezoidal rule using two strips, 2

(iii) Simpson's rule using three function values. 2

- (iv) By means of a sketch or otherwise, explain why part (iii) gives a closer approximation to part (i) than part (ii). 2

(c)



Find the exact volume of a metal component of a yacht's rigging system that is made in the shape of the solid formed when the curve $y = 2(e^{0.5x} + e^{-0.5x})$ between $x = -1$ and $x = 1$ is revolved around the x -axis. 5

Section C

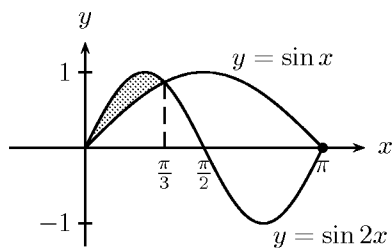
(Use a separate writing booklet.)

Marks

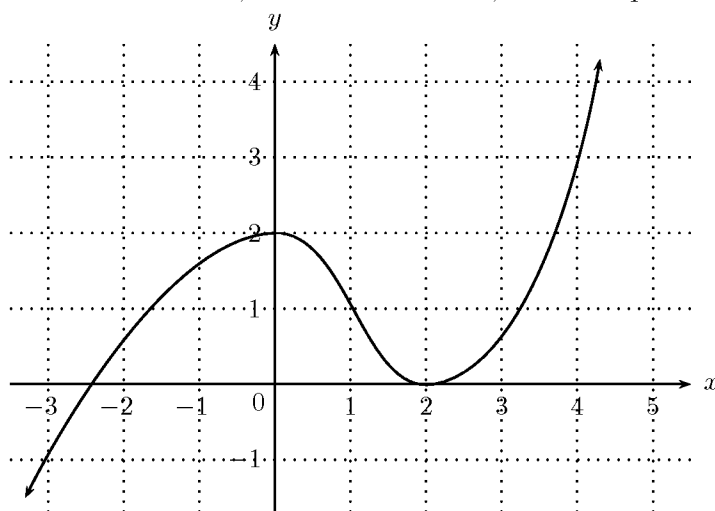
Question 5 (15 marks)

- (a) (i) Sketch the region bounded by $y = 1$, $x = 8$, and the curve $y = \frac{1}{x}$. 2
- (ii) Determine the area of this region with the aid of an appropriate integral. 2

- (b) Calculate the area of the shaded region in the sketch below. 3



- (c) The curve shown is a differential function, $y = f'(x)$. Copy the curve onto your answer booklet and, on the same axes, sketch a possible $y = f(x)$. 4



- (d) Given that $10^x = e^{x \ln 10}$, find

(i) $\frac{d}{dx}(10^x)$, 2

(ii) $\int 10^x dx$. 2

Question 6 (15 marks)

- (a) A hemispherical bowl of radius r cm is to be partly filled with water to a depth of h cm.

(i) What is the domain of h ?

1

(ii) Draw a clear diagram with an appropriately labelled set of axes.

1

(iii) By using the method of volume of revolution, find the volume of water when the depth is h .

3

- (b) (i) Differentiate $x \ln x$ and express in simplest form.

1

(ii) Hence find $\int \ln x \, dx$.

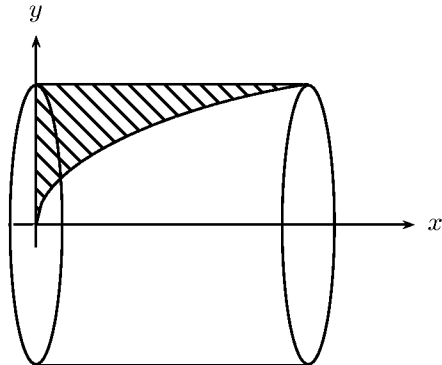
2

- (c) The gradient of a curve at any point on it is $\frac{2}{2x+1}$ and the curve passes through the point $(1, \log_e 3)$. Find the equation of the curve.

3

- (d) The area bounded by the curve $y = \sqrt{\sin x}$, the line $y = \frac{1}{\sqrt{2}}$, and the y -axis is rotated about the x -axis. Find the exact volume of this solid of revolution.

4



End of Paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x$, $x > 0$

June 2008 2nd Maths Assess Task 3.

Section A

1 (a) (i) $\int (2x-3) dx$

$$= \frac{2x^2}{2} - 3x + C$$

$$= x^2 - 3x + C \quad (1)$$

(ii) $\int (x^2 - 2x + 1) dx$

$$= \frac{x^3}{3} - \frac{2x^2}{2} + 1x + C$$

$$= \frac{x^3}{3} - x^2 + x + C \quad (1)$$

(iii) $\int 0 dx$

$$= C \quad (\text{where } C \text{ is a constant}) \quad (1)$$

(Writing a number is too specific ($\frac{1}{2}$ mark).)

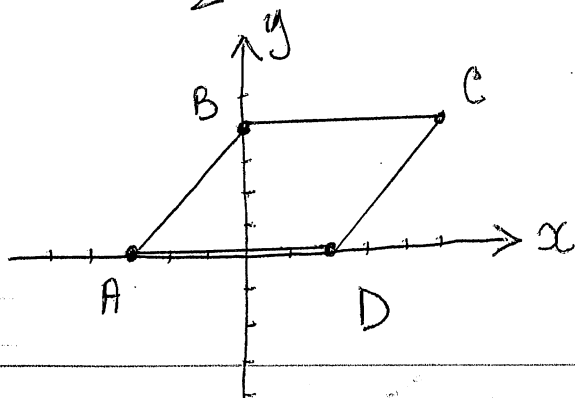
(iv) $\int (3x+1)(x-2) dx$

$$= \int (3x^2 - 5x - 2) dx$$

$$= \frac{3x^3}{3} - \frac{5x^2}{2} - 2x + C$$

$$= x^3 - \frac{5x^2}{2} - 2x + C \quad (2)$$

(b)



$$\text{gradient BC} = \frac{4-4}{5-0} = 0$$

$$\text{gradient AD} = \frac{0-0}{2-3} = 0$$

$BC \parallel AD$

$$\text{gradient AB} = \frac{4-0}{0-3} = -\frac{4}{3}$$

$$\text{gradient CD} = \frac{0-4}{2-5} = \frac{4}{3}$$

$AB \parallel CD$

$$\text{distance } BC = \sqrt{(5-0)^2 + (4-4)^2} = 5$$

$$\text{distance } AB = \sqrt{(0-3)^2 + (4-0)^2} = 5$$

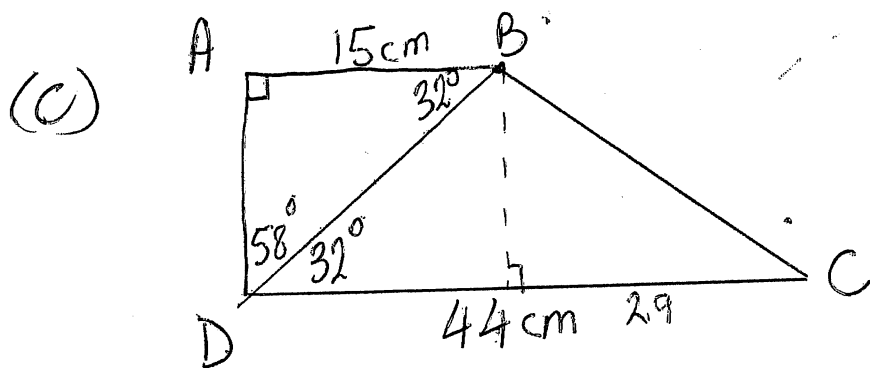
So ABCD could be a rhombus.

(3)

(ii) midpt AC (1, 2). } diagonals intersect each other.
midpt BD (1, 2). }

Gradient AC is $\frac{4-0}{5-3} = \frac{1}{2}$. } since gradients $\times -1$.

Gradient BD is $\frac{0-4}{2-0} = -2$. } they are $\perp$. (2)



(i) Is $AB \parallel CD$?

Yes, $\hat{ABD} = \hat{BDC} = 32^\circ$
angles in alternate position.

(2)

(ii) $\sin 58^\circ = \frac{15}{DB}$

$$DB = \frac{15}{\sin 58^\circ} \approx 17.688 \text{ (3DP)}$$

Now $(BC)^2 = (17.688)^2 + 44^2 - 2 \times 17.688 \times 44 \times \cos 32^\circ$
 $BC \approx 30.477 \text{ 3DP}$

(3)

2 (a) $L = r\theta$

$$L = 40 \times \frac{25\pi}{180}$$

$$\approx 17.5 \text{ cm.}$$

$$180^\circ = \pi$$

$$1^\circ = \frac{\pi}{180}$$

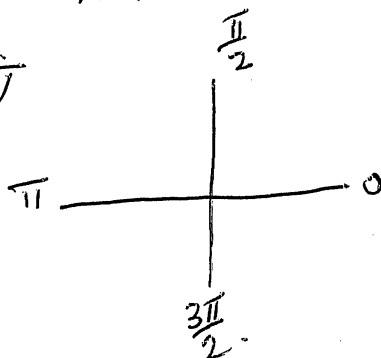
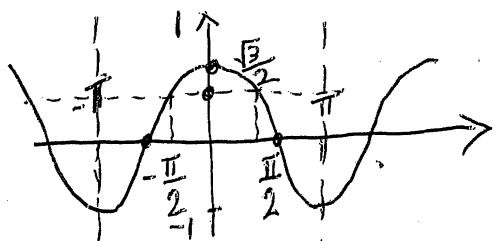
$$25^\circ = \frac{25\pi}{180}$$

(2)

(b) (i) $\sin \frac{\pi}{4} = \sin 45^\circ = \frac{1}{\sqrt{2}}$ (1)

(ii) $\cot \frac{4\pi}{3} = \cot 240^\circ = \frac{1}{\tan 240^\circ} = \frac{1}{\tan 60^\circ} = \frac{1}{\sqrt{3}}$ (1)

(iii) $\cos \theta = \frac{\sqrt{3}}{2}, -\pi \leq \theta \leq \pi$



$\theta = \frac{\pi}{6} \text{ and } -\frac{\pi}{6}$ (2)

(c) (i) $\cos(3x+\pi) + 3 = 3\cos(3x+\pi)$ (1)

(ii) $\frac{1}{x^2-x+2} \times 2x-1 = \frac{2x-1}{x^2-x+2}$ (1)

(iii) $2e^{2x} - e^x + e^{-x}$ (1)

(iv) $\frac{1}{\tan \theta} \times \sec^2 \theta = \frac{\sec^2 \theta}{\tan \theta}$ or $\frac{\frac{1}{\cos^2 \theta}}{\frac{\sin \theta}{\cos \theta}} = \frac{1}{\cos \theta} \times \frac{\cos \theta}{\sin \theta}$

(d) (i) $\sin 2 \approx 0.91$ (1)

(ii) $\ln 17 \approx 2.83$ (1)

(e) let $y = e^{x^2}$

$y' = 2x e^{x^2}$

$u = 1x \dots 1x \dots x^2 \dots 1x \dots x^2 \dots 4x^2 \dots x^2 \dots 1x^2 \dots$

(2)

Section B

Question 3

a) i $\int_0^{\pi/4} \cos 2x \, dx = 0.5$

LHS = $\int_0^{\pi/4} \cos 2x \, dx$

$$= \left[\frac{1}{2} \sin 2x \right]_0^{\pi/4}$$

$$= \frac{1}{2} \sin \frac{2\pi}{4} - \frac{1}{2} \sin 0$$

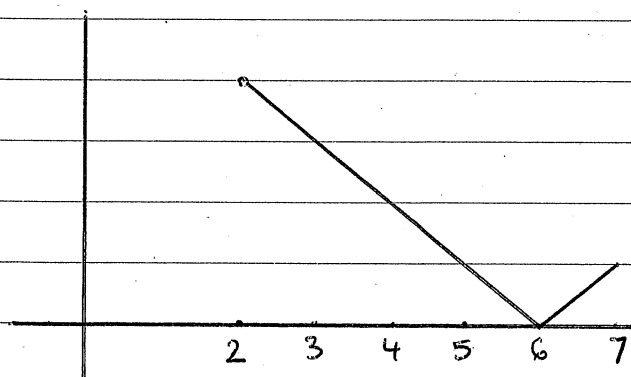
$$= \frac{1}{2} - 0$$

$$= \frac{1}{2} = \text{RHS.} \quad (2)$$

ii $\int_2^7 \left| 3 - \frac{x}{2} \right| dx = 4.25$

x	2	3	4	5	6	7
$f(x)$	2	$1\frac{1}{2}$	1	$\frac{1}{2}$	0	$\frac{1}{2}$

$\therefore f(x) = \left| 3 - \frac{x}{2} \right| \quad 2 \leq x \leq 7$
looks like



$\therefore \int_2^7 \left| 3 - \frac{x}{2} \right| dx = \text{area of big } \Delta + \text{area of small } \Delta$

$$A = \frac{1}{2} \times 4 \times 2 + \frac{1}{2} \times 1 \times \frac{1}{2}$$

$$= 4.25$$

OR

$$\int_2^6 \frac{3-x}{2} dx + \int_6^7 -\left(3 - \frac{x}{2}\right) dx$$

$$= \left[3x - \frac{x^2}{4} \right]_2^6 + \left[\frac{x^2}{4} - 3x \right]_6^7$$

$$= \left(18 - \frac{36}{4} \right) - \left(6 - \frac{4}{4} \right) + \left(\frac{49}{4} - 21 \right)$$

$$= \left(\frac{36}{4} - 18 \right)$$

$$= 9 - 5 + \frac{-35}{4} + 9$$

$$= 4.25$$

(3)

b) Circle through $(3,0)$ & $(12,0)$, tangent to y axis.

∴ distances from centre of circle to $(3,0)$, $(12,0)$ or $(0,y)$ - tangent. are equal - radii.
let Centre = (x, y)

D_o A to Centre = D_o B to Centre.

$$\sqrt{(3-x)^2 + (0-y)^2} = \sqrt{(12-x)^2 + (0-y)^2}$$

$$(3-x)^2 + (-y)^2 = (12-x)^2 + (-y)^2$$

$$9 - 6x + x^2 + y^2 = 144 - 24x + x^2 + y^2$$

$$18x = 135$$

$$\therefore x = 7.5$$

D_o A to Centre = D_o Tangent to Cent

The y, value must be equal to the y value of the tangent.

tangent to circle makes $\perp$ with radii.

$$\therefore \sqrt{(3-7.5)^2 + (0-y)^2} = \sqrt{(7.5-0)^2 + (y-0)^2}$$

$$\therefore \sqrt{(-4.5)^2 + (-y)^2} = \sqrt{(7.5)^2 + 0^2}$$

$$(-4.5)^2 + y^2 = 7.5^2$$

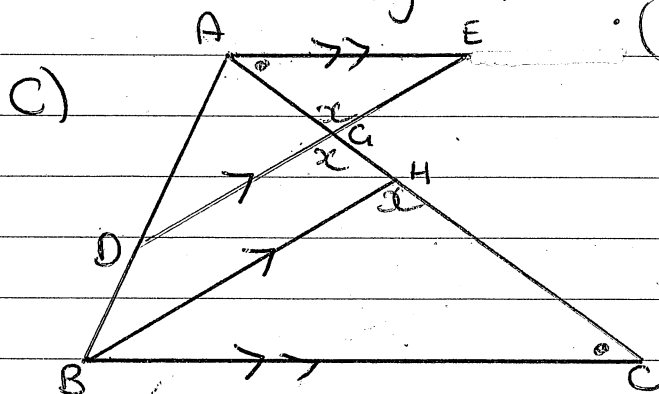
$$y^2 = 7.5^2 - (-4.5)^2 = 6$$

∴ Centre of circle is $(7.5, \pm 6)$

$$\text{radius} = 7.5$$

∴ eqn of circle

$$(x-7.5)^2 + (y-6)^2 = 56.25 \quad (4)$$



i) let $\angle AEG = x$ (3)

∴ $\angle DGH = x$ (vert opp)

∴ $\angle BHC = x$ (corresponding) (3)

ii) $\angle AEG = \angle ACB$ (alt L's)

$\angle BHC = \angle AGE$ (as above)

∴ $\angle AEG = \angle CBH$ (L sum Δ)

∴ $\Delta AEG \parallel \Delta CBH$

Question 4

a) $\sin x = 0.8$

$\therefore x = \sin^{-1} 0.8$

$0 \leq x \leq 2\pi$

$\therefore x = 0.9273, 2.2143$

b) i) $\int_0^{\pi/4} \sec^2 x \, dx$
 $= [\tan x]_0^{\pi/4}$
 $= 1 - 0$
 $= 1$

ii) $\int_0^{\pi/4} \sec^2 x \, dx$

$\therefore \frac{h}{2} [y_0 + y_n] + 2[y_1]$

$h = \frac{b-a}{n} = \frac{\pi/4 - 0}{8} = \frac{\pi}{8}$

$\therefore \frac{\pi/8}{2} \left[\sec^2 0 + \sec^2 \frac{\pi}{4} \right] + 2 \left[\sec^2 \frac{\pi}{8} \right]$

$= \frac{\pi}{16} [1 + 2] + 2 \cdot 2.43145751$

$= 1.0492124215$

iii) $\int_0^{\pi/4} \sec^2 x \, dx$

$\therefore \frac{b-a}{6} \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right]$

$= \frac{\pi/4 - 0}{6} \left[\sec^2 0 + 4\sec^2 \frac{\pi}{8} + \sec^2 \frac{\pi}{4} \right]$

$= \frac{\pi}{24} [1 + 4 \cdot 6.86291501 + 2]$

$= 1.006133205$

iv) Simpsons rule is generally more accurate as it uses parabolic arcs instead of straight lines.

c) $V = \pi \int_{-1}^1 [2(e^{0.5x} + e^{-0.5x})]^2 dx$

$= \pi \int_{-1}^1 4(e^x + e^{-x} + 2) dx$

$= 4\pi \left[\frac{e^x}{1} + \frac{e^{-x}}{-1} + 2x \right]_{-1}^1$

$= 4\pi \left[e^x - e^{-x} + 2x \right]_{-1}^1$

$= 4\pi \left[\left(e - \frac{1}{e} + 2 \right) - \left(\frac{1}{e} - e - 2 \right) \right]$

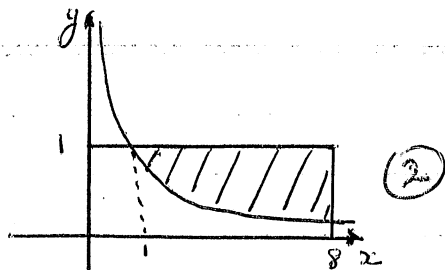
$= 4\pi \left[\frac{e-1}{e} + 2 - \frac{1-e}{e} + 2 \right]$

$= 4\pi \left[2e + 4 - \frac{2}{e} \right]$

$= 8\pi \left(e + 2 - \frac{1}{e} \right)$

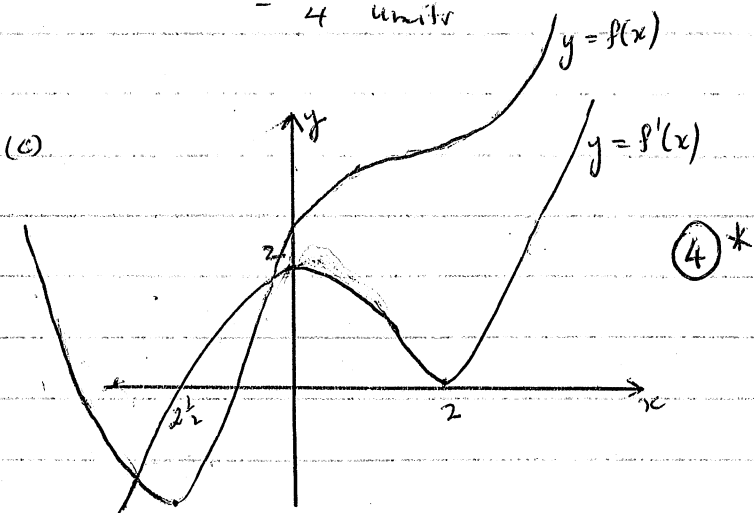
Question 5:

(a) (i)



$$\begin{aligned}
 \text{(ii) Area} &= 7 \times 1 - \int_1^8 \frac{1}{x} dx \\
 &= 7 - [\ln x]_1^8 \\
 &= 7 - \{\ln 8 - \ln 1\} \\
 &= 7 - \ln 8 \quad \text{units}^2 \quad (2) \\
 &(\approx 4.920558458 \dots)
 \end{aligned}$$

$$\begin{aligned}
 \text{(b) Area} &= \int_0^{\pi/3} (\sin 2x - \sin x) dx \\
 &= \left[-\frac{1}{2} \cos 2x + \cos x \right]_0^{\pi/3} \\
 &= \left[-\frac{1}{2} \cos \frac{2\pi}{3} + \cos \frac{\pi}{3} \right] - \left[-\frac{1}{2} \cos 0 + \cos 0 \right] \\
 &= \left[-\frac{1}{2} \times -\frac{1}{2} + \frac{1}{2} \right] - \left[-\frac{1}{2} + 1 \right] \\
 &= \frac{3}{4} - \frac{1}{2} \\
 &= \frac{1}{4} \quad \text{units}^2 \quad (3)
 \end{aligned}$$



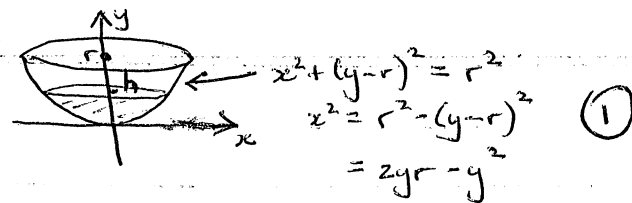
$$\begin{aligned}
 \text{(d) (i) } \frac{d}{dx} (10^x) &= \frac{d}{dx} (e^{x \ln 10}) \\
 &= \ln 10 \cdot e^{x \ln 10} \\
 &= \ln 10 \cdot 10^x \quad (2)
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii) } \int 10^x dx &= \int e^{x \ln 10} dx \\
 &= \frac{1}{\ln 10} e^{x \ln 10} + C \\
 &= \frac{1}{\ln 10} 10^x + C \quad (2)
 \end{aligned}$$

15

Question 6:

(a) (ii)



$$(i) \quad 0 \leq h \leq r \quad (1)$$

$$\begin{aligned}
 \text{(ii) Volume} &= \pi \int_0^h x^2 dy \\
 &= \pi \int_0^h (2yr - y^2) dy \\
 &= \pi \left[y^2 r - \frac{1}{3} y^3 \right]_0^h \\
 &= \pi \left[rh^2 - \frac{h^3}{3} \right] - 0 \\
 &= \pi h^2 \left(r - \frac{h}{3} \right) \\
 &= \frac{\pi h^2}{3} (3r - h) \quad (3)
 \end{aligned}$$

$$\begin{aligned}
 \text{(b) (i) } \frac{d}{dx} (x \ln x) &= \ln x \cdot 1 + x \cdot \frac{1}{x} \\
 &= \ln x + 1 \quad (1)
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii) } \therefore \int \frac{d}{dx} (x \ln x) dx &= \int (\ln x + 1) dx \\
 \therefore x \ln x &= \int \ln x dx + x + C \\
 \therefore \int \ln x dx &= x \ln x - x + C, \quad (2)
 \end{aligned}$$

$$\begin{aligned}
 \text{(c) } y' &= \frac{2}{2x+1} \\
 \therefore y &= \ln(2x+1) + C \\
 \text{When } x=1: \ln 3 &= \ln 3 + C \\
 \therefore C &= 0 \\
 \therefore y &= \ln(2x+1) \quad (3)
 \end{aligned}$$

$$\begin{aligned}
 \text{(d) Volume} &= \pi \left(\frac{1}{\sqrt{2}} \right)^2 \cdot \frac{\pi}{6} \\
 &= \frac{\pi^2}{12} - \pi \int_0^{\pi/6} (\sin x)^2 dx \\
 &= \frac{\pi^2}{12} + \pi \left[\cos x \right]_0^{\pi/6} \\
 &= \frac{\pi^2}{12} + \pi \left[\frac{\sqrt{3}}{2} - 1 \right] \quad \text{units}^3 \quad (4)
 \end{aligned}$$

15